Bharath Mandha

AI/ML Engineer, 4+ years in production ML; RAG and LLM evaluation at Goldman Sachs
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bharathkumardev1.github.io (live demos)

EXPERIENCE

Goldman Sachs, AI/ML Engineer — Jun 2025 to Present

- Co-designed and optimized the retrieval layer (metadata filtering, reranking) and the LangGraph orchestration layer, and built their FastAPI integration, for a RAG platform over 3M+ internal financial documents (tens of millions of chunks) used weekly by 70+ credit and risk analysts, working with platform, risk and compliance teams
- The platform's automated evidence discovery, document analysis and validation, with analyst review preserved, is reported to have cut average credit-memo preparation time by about 40% (project-level estimate, not my own time study)
- Built a 500+ query RAG evaluation benchmark (factual retrieval, financial comparisons, evidence-based reasoning) scoring retrieval with Hit Rate and MRR and generation with RAGAS faithfulness, groundedness and answer relevance (end-to-end workflow: 10 to 20 s); used it to compare chunk size, overlap, top-k and embedding choices experiment by experiment, alongside metadata-filtering and index-performance work; groundedness above 90%
- Built LangGraph agents for retrieval, analysis, validation and delegation that took over repetitive credit-analysis reasoning steps, with analysts reviewing sensitive decisions
- Implemented document preprocessing, recursive and semantic chunking, metadata enrichment, embedding generation and incremental FAISS index updates in the team's PySpark and AWS S3 ingestion pipeline, so new or modified documents reached the index without a full rebuild for routine updates
- Evaluated quantization and LoRA/QLoRA approaches for open-source models to assess GPU-memory requirements, inference latency, model quality and serving cost, and concluded that infrastructure savings alone didn't justify moving complex credit reasoning to smaller models

Cigna Healthcare, Machine Learning Engineer — Mar 2022 to Aug 2023

- Built 30-day hospital readmission-risk prediction with XGBoost and LightGBM over 1.5M+ HIPAA-regulated encounters, reaching 0.91 ROC-AUC on a held-out cohort of 200K+ encounters from 70K+ patients with time-aware, patient-level splits
- Engineered 150+ candidate features from labs, demographics, encounter history and clinical notes; about 100 kept in production for predictive value and stability; handled class imbalance and tuned the decision threshold
- Productionized training, batch inference, registration and retraining (MLflow, Airflow) on SageMaker and EKS through CI/CD with PSI drift monitoring; SHAP explanations let clinical reviewers prioritize several thousand high-risk patients monthly

Best Buy, Associate Data Scientist — Jun 2020 to Feb 2022

- Cut weighted MAPE of SKU-and store-level demand forecasts by about 14% relative with gradient-boosted models benchmarked against time-series baselines on time-based holdouts, tracking RMSE alongside MAPE
- Built PySpark, SQL and BigQuery pipelines over 50M+ transactions across thousands of SKUs and hundreds of stores
- Raised response on comparable campaigns by about 12% with six customer segments from clustering and cohort analysis

PROJECTS (CODE, RESULTS AND CI PUBLIC ON GITHUB)

finance-rag-eval, retrieval over SEC filings — 2026

- Page-level retrieval evaluation on FinanceBench (150 questions, 360 filings, 53,399 pages) with a held-out split and paired bootstrap intervals: a company and period filter (+0.29) and statement-page routing (+0.28) take Hit@10 from 0.14 to 0.71, and leaving the company name out of the query reaches 0.75 (within noise)
- A MiniLM hybrid led on dev but tied BM25 on test (0.73 vs 0.75) at about 5x the latency, so the service runs BM25: 126 queries/s per core at 12 ms p95, an estimated $0.10 per million queries (not measured); a browser port matches its scores bit for bit

readmission-capacity, outreach planning on public hospital data — 2026

- Readmission models judged at a fixed call capacity on 101,766 UCI encounters split by patient: a 10% LightGBM call list holds 25.0% of 30-day readmissions (95% CI 23.2 to 26.7%) vs 20.5% for prior stays and 15.8% for LACE, with 539 readmissions expected on the list and 569 observed
- TreeSHAP reason codes with code descriptions checked in CI against the dataset's code table; simulated shifts caught by PSI and missing-value alerts; CI re-downloads the data, retrains and fails if headline metrics move

SKILLS

Languages and data: Python, SQL, PySpark, BigQuery

ML and LLMs: PyTorch, scikit-learn, XGBoost, LightGBM, SHAP, Hugging Face, LoRA/QLoRA, LangGraph, LangChain, FAISS, RAGAS

MLOps and cloud: MLflow, Airflow, FastAPI, Docker, Kubernetes, Terraform, CI/CD, AWS (SageMaker, S3, EKS), GCP Vertex AI

EDUCATION

University of North Texas, Master of Science in Computer Science — Aug 2023 to May 2025